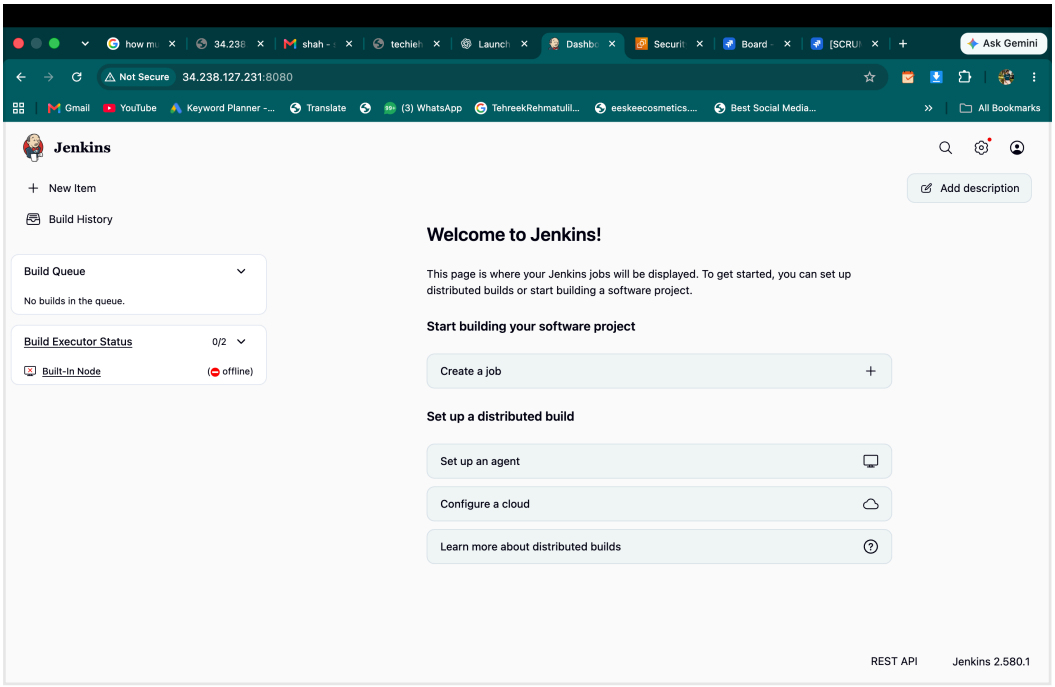
1.Launch one aws ec2 and install jenkins.

1.Launch an AWS EC2 instance with Amazon Linux 2023, create a key pair, and configure inbound rules for SSH (22) and Jenkins (8080).

2.Connect to EC2 using SSH, update the system, and install Java 21 using sudo dnf install fontconfig java-21-amazon-corretto -y.

3.Add the official Jenkins repository, install Jenkins using sudo dnf install jenkins -y, and start it with sudo systemctl enable --now jenkins.

4.Open http://YOUR_EC2_PUBLIC_IP:8080, retrieve the initial password using sudo cat /var/lib/jenkins/secrets/initialAdminPassword, and complete the Jenkins setup.



```
Last login: Mon Oct 12 15:54:22 on ttys001
sharafkamaal@192 ~ % ssh ec2-user@34.238.127.231
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 12 07:12:02 2026 from 103.203.172.230
[ec2-user@ip-172-31-36-3 ~]$ ^[[200~sudo apt update
-bash: $'\E[200~sudo': command not found
[ec2-user@ip-172-31-36-3 ~]$ sudo apt install fontconfig openjdk-21-jre -y
sudo: apt: command not found
[ec2-user@ip-172-31-36-3 ~]$ sudo apt update
sudo apt install fontconfig openjdk-21-jre -y
java -version
sudo: apt: command not found
sudo: apt: command not found
-bash: java: command not found
[ec2-user@ip-172-31-36-3 ~]$ ^C
[ec2-user@ip-172-31-36-3 ~]$ sudo dnf update -y
Amazon Linux 2023 Kernel Livepatch repository                                                    717 kB/s |  89 kB     00:00
Last metadata expiration check: 0:00:01 ago on Tue Oct 13 07:40:56 2026.
================================================================================================================================
WARNING:
  A newer release of "Amazon Linux" is available.
```

```
[ec2-user@ip-172-31-36-3 ~]$ sudo apt update
sudo apt install fontconfig openjdk-21-jre -y
java -version
sudo: apt: command not found
sudo: apt: command not found
-bash: java: command not found
[ec2-user@ip-172-31-36-3 ~]$ ^C
[ec2-user@ip-172-31-36-3 ~]$ sudo dnf update -y
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:01 ago on Tue Oct 13 07:40:56 2026.
================================================================================
```

```
[ec2-user@ip-172-31-36-3 ~]$ java -version
openjdk version "21.0.12.1" 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS, mixed mode, sharing)
[ec2-user@ip-172-31-36-3 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-10-13 07:43:45--  https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo 100%[=================================================>]     267  --.-KB/s    in 0s

2026-10-13 07:43:45 (2.74 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]
```

```
[ec2-user@ip-172-31-36-3 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-10-13 07:44:39 UTC; 13min ago
   Main PID: 151848 (java)
      Tasks: 37 (limit: 1059)
     Memory: 357.6M
        CPU: 20.879s
     CGroup: /system.slice/jenkins.service
             └─151848 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot

Oct 13 07:44:34 ip-172-31-36-3.ec2.internal jenkins[151848]: [LF]>
```

3.Setup Apache Tomcat on Ubuntu Virtual Machine

- Apache Tomcat was installed on Ubuntu using the following

commands:

- sudo apt update
- sudo apt install openjdk-21-jdk -y
- **Tomcat was downloaded, extracted to /opt/tomcat, and configured with the correct permissions:**
- sudo tar -xzf /tmp/apache-tomcat-11.0.27.tar.gz -C /opt/tomcat --strip-components=1
- sudo chown -R tomcat:tomcat /opt/tomcat
- Tomcat was configured as a systemd service and verified using:
- sudo systemctl enable tomcat
- sudo systemctl start tomcat
- sudo systemctl status tomcat
- curl -I http://localhost:8080

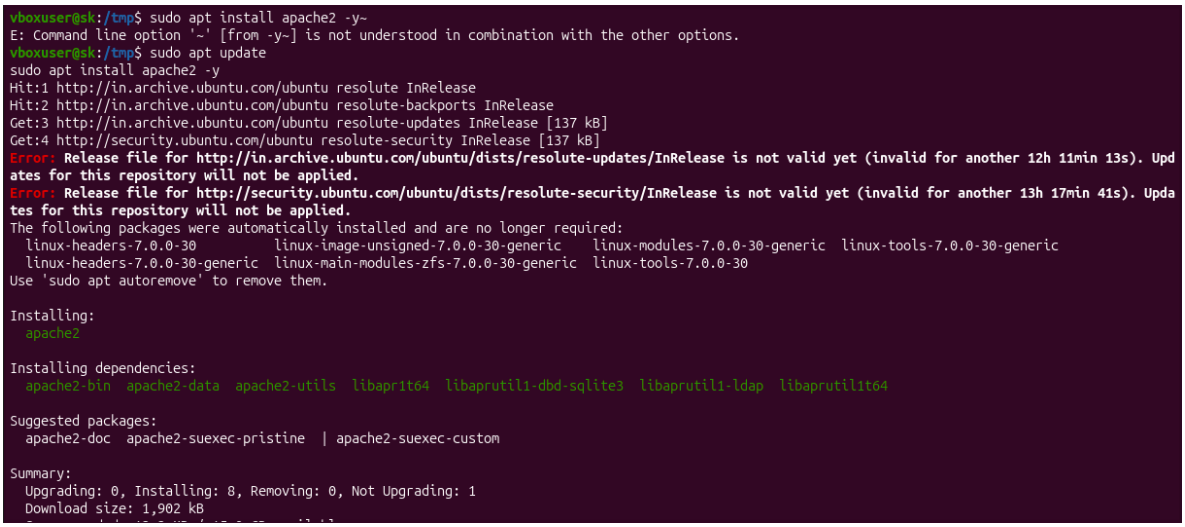
```
vboxuser@sk:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Error: Release file for http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease
tes for this repository will not be applied.
Error: Release file for http://in.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease
tes for this repository will not be applied.
vboxuser@sk:~$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30                  linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic          linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic   linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.
```

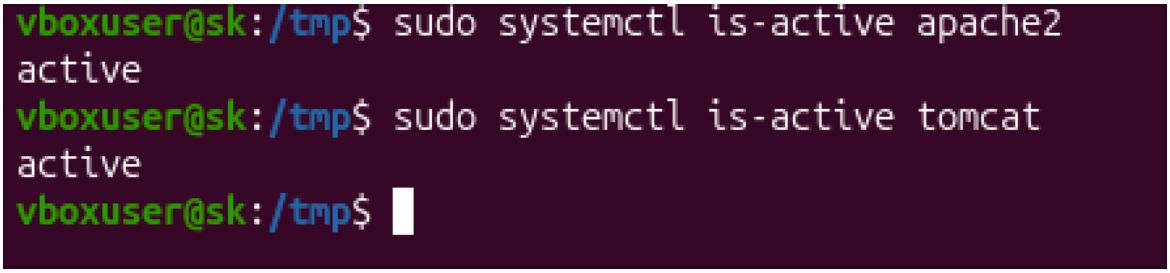
```
vboxuser@sk:~$ java -version
javac -version
openjdk version "21.0.12.1" 2026-08-18
OpenJDK Runtime Environment (build 21.0.12.1+1-1-26.04.4-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12.1+1-1-26.04.4-Ubuntu, mixed mode, sharing)
javac 21.0.12.1
vboxuser@sk:~$ sudo useradd --system --home-dir /opt/tomcat \
  --shell /usr/sbin/nologin tomcat
vboxuser@sk:~$
vboxuser@sk:~$ sudo useradd --system --home-dir /opt/tomcat \                    --shell /usr/sbin/nologin tomcat^C

vboxuser@sk:~$ cd /tmp
vboxuser@sk:/tmp$ wget https://downloads.apache.org/tomcat/tomcat-11/v11.0.27/bin/apache-tomcat-11.0.27.tar.gz
--2026-10-12 13:07:48--  https://downloads.apache.org/tomcat/tomcat-11/v11.0.27/bin/apache-tomcat-11.0.27.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 95.216.224.44, 2a01:4f9:2b:1cc2::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14545574 (14M) [application/x-gzip]
Saving to: ‘apache-tomcat-11.0.27.tar.gz’

apache-tomcat-11.0. 100%[===================>]  13.87M   730KB/s    in 2m 7s

2026-10-12 13:09:55 (112 KB/s) - ‘apache-tomcat-11.0.27.tar.gz’ saved [14545574/14545574]
```





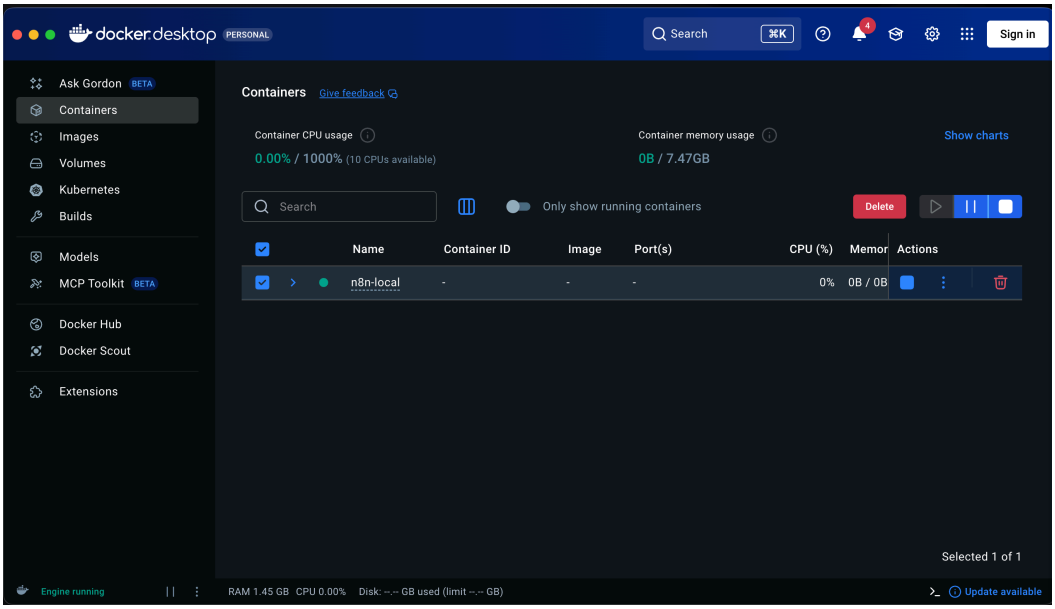
3.Setup Docker Desktop on your local windows.

1. Download Docker Desktop for Windows from the official Docker website.

2. Run the installer, select Use WSL 2 instead of Hyper-V, and complete the installation. Restart Windows if prompted.

3. Open PowerShell as Administrator and run wsl --install if WSL 2 is not installed. Ensure hardware virtualization is enabled in BIOS/UEFI.

4. Launch Docker Desktop and verify the installation in PowerShell using docker --version and docker run hello-world.


docker desktop
PERSONAL
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Containers
Give feedback
Container CPU usage
0.00% / 1000% (10 CPUs available)
Container memory usage
0B / 7.47GB
Show charts
Search
Only show running containers
Delete
Name
Container ID
Image
Port(s)
CPU (%)
Memor
Actions
n8n-local
0%
0B / 0B
Selected 1 of 1
Engine running
RAM 1.45 GB CPU 0.00% Disk: --.-- GB used (limit --.-- GB)
Update available